

Change the title of Clause 124 as follows:

124. Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2

124.1 Overview

Change the first paragraph in 124.1 as follows:

This clause specifies the 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2 PMDs together with the single-mode fiber medium. The optical signal generated by this PMD type is modulated using a 4-level pulse amplitude modulation (PAM4) format. When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 124–1 for 400GBASE-DR4 and 400GBASE-DR4-2 and in Table 124–1a for 800GBASE-DR8 and 800GBASE-DR8-2, to the medium through the MDI and optionally with the management functions that may be accessible through the management interface defined in Clause 45, or equivalent.

Change Table 124–1 (as amended by IEEE Std 802.3ck-2022) as follows:

**Table 124–1—Physical Layer clauses associated with the 400GBASE-DR4 and
400GBASE-DR4-2 PMDs**

Associated clause	<u>400GBASE-DR4, 400GBASE-DR4-2</u>
117—RS	Required
117—400GMII ^a	Optional
118—400GMII Extender	Optional
119—PCS	Required
120—PMA	Required
120B—Chip-to-chip 400GAUI-16	Optional
120C—Chip-to-module 400GAUI-16	Optional
120D—Chip-to-chip 400GAUI-8	Optional
120E—Chip-to-module 400GAUI-8	Optional
120F—Chip-to-chip 400GAUI-4 (400GAUI-4 C2C)	Optional
120G—Chip-to-module 400GAUI-4 (400GAUI-4 C2M)	Optional
78—Energy Efficient Ethernet	Optional

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

Insert new Table 124–1a after Table 124–1 as follows:

Table 124–1a—Physical Layer clauses associated with the 800GBASE-DR8 and 800GBASE-DR8-2 PMDs

Associated clause	800GBASE-DR8, 800GBASE-DR8-2
170—RS	Required
170—800GMII ^a	Optional
171—800GMII Extender	Optional
172—PCS	Required
173—PMA	Required
120F—Chip-to-chip 800GAUI-8	Optional
120G—Chip-to-module 800GAUI-8	Optional

^a The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

Change the second paragraph in 124.1 as follows:

Figure 124–1 shows the relationship of the PMD and MDI (shown shaded) with other sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 400 Gb/s Ethernet is introduced in Clause 116 and the purpose of each PHY sublayer is summarized in 116.2. 800 Gb/s Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.

Insert the following new paragraph before the last paragraph in 124.1:

400GBASE-DR4 and 400GBASE-DR4-2 use four lanes, while 800GBASE-DR8 and 800GBASE-DR8-2 use eight lanes. In this clause, where there are four or eight items (depending on PMD type) such as lanes, the items are numbered from 0 to $n - 1$, where n is 4 or 8, and an example item is numbered i .

Replace Figure 124–1 with the following figure:

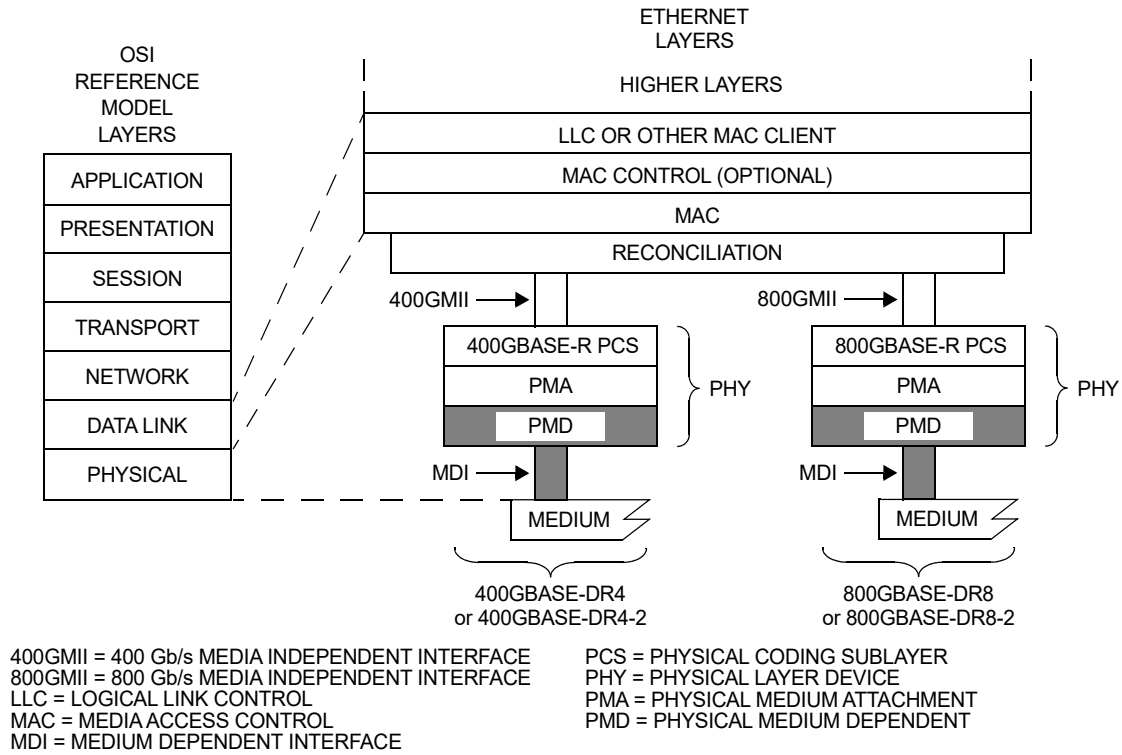


Figure 124–1—400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

Change the last paragraph in 124.1 as follows:

400GBASE-DR4 and 400GBASE-DR4-2 PHYs with the optional Energy Efficient Ethernet (EEE) fast wake capability may enter the Low Power Idle (LPI) mode to conserve energy during periods of low link utilization (see Clause 78). The deep sleep mode of EEE is not supported.

124.1.1 Bit error ratio

Change 124.1.1 as follows:

The bit error ratio (BER) for 400GBASE-DR4 and 400GBASE-DR4-2 PMDs, when processed according to Clause 120, shall be less than 2.4×10^{-4} provided that the error statistics are sufficiently random that this results in a frame loss ratio (see 1.4.344) of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap when processed according to Clause 120 and then Clause 119. ~~For a complete Physical Layer, the frame loss ratio may be degraded to 6.2×10^{-11} for 64-octet frames with minimum interpacket gap due to additional errors from the electrical interfaces. If the error statistics are not sufficiently random to meet this requirement, then the BER shall be less than that required to give a frame loss ratio of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap.~~

The bit error ratio (BER) for 800GBASE-DR8 and 800GBASE-DR8-2 PMDs, when processed according to Clause 173, shall be less than 2.4×10^{-4} provided that the error statistics are sufficiently random that this results in a frame loss ratio (see 1.4.344) of less than 3.4×10^{-12} for 64-octet frames with minimum

interpacket gap when processed according to Clause 173 and then Clause 172. If the error statistics are not sufficiently random to meet this requirement, then the BER shall be less than that required to give a frame loss ratio of less than 3.4×10^{-12} for 64-octet frames with minimum interpacket gap.

For a complete Physical Layer, the frame loss ratio may be degraded to 6.2×10^{-11} for 64-octet frames with minimum interpacket gap due to additional errors from the electrical interfaces.

~~If the error statistics are not sufficiently random to meet this requirement, then the BER shall be less than that required to give a frame loss ratio of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap.~~

124.2 Physical Medium Dependent (PMD) service interface

Change the first six paragraphs in 124.2 as follows:

This subclause specifies the services provided by the 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2 PMDs. The service interface for ~~this~~ these PMDs is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD entity. The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3 for the 400GBASE-DR4 and 400GBASE-DR4-2 PMDs and in 169.3 for the 800GBASE-DR8 and 800GBASE-DR8-2 PMDs. The PMD service interface primitives are summarized as follows:

PMD:IS_UNITDATA_ *i*.request
PMD:IS_UNITDATA_ *i*.indication
PMD:IS_SIGNAL.indication

The 400GBASE-DR4 and 400GBASE-DR4-2 PMDs ~~has~~ have four parallel symbol streams, hence $i = 0$ to 3. The 800GBASE-DR8 and 800GBASE-DR8-2 PMDs have eight parallel symbol streams, hence $i = 0$ to 7.

In the transmit direction, the PMA continuously sends ~~four~~ n parallel symbol streams to the PMD, one per lane, each at a nominal signaling rate of 53.125 GBd. The PMD then converts these streams of data units into the appropriate signals on the MDI.

In the receive direction, the PMD continuously sends ~~four~~ n parallel symbol streams to the PMA corresponding to the signals received from the MDI, one per lane, each at a nominal signaling rate of 53.125 GBd. ~~See For 400GBASE-DR4 and 400GBASE-DR4-2, see~~ NOTE at the end of 120.5.2 concerning the transition density of lanes operating at this nominal signaling rate.

The SIGNAL_DETECT parameter defined in this clause maps to the SIGNAL_OK parameter in the PMD:IS_SIGNAL.indication(SIGNAL_OK) inter-sublayer service interface primitive defined in 116.3 for 400GBASE-DR4 and 400GBASE-DR4-2 and defined in 169.3 for 800GBASE-DR8 and 800GBASE-DR8-2.

124.3 Delay and Skew

124.3.1 Delay constraints

Change 124.3.1 as follows:

The sum of the transmit and receive delays at one end of the link contributed by the 400GBASE-DR4 or 400GBASE-DR4-2 PMD including 2 m of fiber in one direction shall be no more than 8192 bit times (16 pause_quanta or 20.48 ns). ~~A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4 and its references.~~

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-DR8 or 800GBASE-DR8-2 PMD including 2 m of fiber in one direction shall be no more than 16 384 bit times (32 pause_quanta or 20.48 ns).

Descriptions of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4 for 400GBASE-DR4 and 400GBASE-DR4-2, and in 169.4 for 800GBASE-DR8 and 800GBASE-DR8-2.

124.3.2 Skew constraints

Change 124.3.2 as follows:

The Skew (relative delay between the lanes) and Skew Variation are kept within limits so that the information on the lanes can be reassembled by the PCS.

124.3.2.1 Skew constraints for 400GBASE-DR4 and 400GBASE-DR4-2

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in [Figure 116–4](#) and [Figure 116–5](#).

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5. The measurements of Skew and Skew Variation are defined in [86.8.3.1](#).

124.3.2.2 Skew constraints for 800GBASE-DR8 and 800GBASE-DR8-2

Skew and Skew Variation are defined in 169.5 and specified at the points SP1 to SP6 shown in [Figure 169–4](#) and [Figure 169–5](#).

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5. The measurements of Skew and Skew Variation are defined in 86.8.3.1.

124.4 PMD MDIO function mapping

Change Table 124–2 and Table 124–3 as follows:

Table 124–2—MDIO/PMD control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	PMD control variable
Reset	PMA/PMD control 1 register	1.0.15	PMD_reset
Global PMD transmit disable	PMD transmit disable register	1.9.0	PMD_global_transmit_disable
PMD transmit disable 3 $n-1$ to PMD transmit disable 0	PMD transmit disable register	1.9.4 to 1.9.1 1.9. n to 1.9.1	PMD_transmit_disable 3 $n-1$ to PMD_transmit_disable_0

Table 124–3—MDIO/PMD status variable mapping

MDIO status variable	PMA/PMD register name	Register/bit number	PMD status variable
Fault	PMA/PMD status 1 register	1.1.7	PMD_fault
Transmit fault	PMA/PMD status 2 register	1.8.11	PMD_transmit_fault
Receive fault	PMA/PMD status 2 register	1.8.10	PMD_receive_fault
Global PMD receive signal detect	PMD receive signal detect register	1.10.0	PMD_global_signal_detect
PMD receive signal detect 3 $n-1$ to PMD receive signal detect 0	PMD receive signal detect register	1.10.4 to 1.10. n to 1.10.1	PMD_signal_detect 3 $n-1$ to PMD_signal_detect_0

124.5 PMD functional specifications

Change 124.5 as follows:

The 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2 PMDs perform
~~performs~~ the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

124.5.1 PMD block diagram

Change the first paragraph in 124.5.1 as follows:

The PMD block diagram for 400GBASE-DR4 or 400GBASE-DR4-2 is shown in Figure 124-2. The block diagrams for 800GBASE-DR8 and 800GBASE-DR8-2 are equivalent to Figure 124-2, but for eight lanes per direction, instead of four. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 124.8 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 124.11.3). Unless specified otherwise, all receiver measurements and tests defined in 124.8 are made at TP3.

Change the title of Figure 124-2 as follows:

Figure 124-2—Block diagram for 400GBASE-DR4 or 400GBASE-DR4-2 transmit/receive paths

124.5.2 PMD transmit function

Change 124.5.2 as follows:

The PMD Transmit function shall convert the ~~four~~_{*n*} symbol streams requested by the PMD service interface messages PMD:IS_UNITDATA_0.request to PMD:IS_UNITDATA_3~~*n*~~_{*n*}.request into ~~four~~_{*n*} separate optical signals. The ~~four~~_{*n*} optical signals shall then be delivered to the MDI, which contains ~~four~~_{*n*} parallel light paths for transmit, according to the transmit optical specifications in this clause. The highest optical power level in each signal shall correspond to tx_symbol = three and the lowest shall correspond to tx_symbol = zero.

124.5.3 PMD receive function

Change 124.5.3 as follows:

The PMD Receive function shall convert the ~~four~~_{*n*} parallel optical signals received from the MDI into separate symbol streams for delivery to the PMD service interface using the messages PMD:IS_UNITDATA_0.indication to PMD:IS_UNITDATA_3~~*n*~~_{*n*}.indication, all according to the receive optical specifications in this clause. The higher optical power level in each signal shall correspond to rx_symbol = three and the lowest shall correspond to rx_symbol = zero.

124.5.4 PMD global signal detect function

Change the first two paragraphs in 124.5.4 as follows:

The PMD global signal detect function shall report the state of SIGNAL_DETECT via the PMD service interface. The SIGNAL_DETECT parameter is signaled continuously, while the PMD:IS_SIGNAL.indication message is generated when a change in the value of SIGNAL_DETECT occurs. The SIGNAL_DETECT parameter defined in this clause maps to the SIGNAL_OK parameter in the inter-sublayer service interface primitives defined in 116.3 for 400GBASE-DR4 and 400GBASE-DR4-2, and in 169.3 for 800GBASE-DR8 and 800GBASE-DR8-2.

SIGNAL_DETECT shall be a global indicator of the presence of optical signals on all ~~four~~_{*n*} lanes. The value of the SIGNAL_DETECT parameter shall be generated according to the conditions defined in Table 124-4. The PMD receiver is not required to verify whether a compliant

~~400GBASE-DR4~~400GBASE-R or 800GBASE-R signal is being received. This standard imposes no response time requirements on the generation of the SIGNAL_DETECT parameter.

Change Table 124–4 as follows:

Table 124–4—SIGNAL_DETECT value definition

Receive conditions	SIGNAL_DETECT value
For any lane; Average optical power at TP3 ≤ -15 dBm	FAIL
For all lanes; [(Optical power at TP3 $\geq$ average receive power, each lane (min) <u>in</u> Table 124–7) AND (compliant 400GBASE-R <u>or</u> 800GBASE-R signal input)]	OK
All other conditions	Unspecified

124.5.5 PMD lane-by-lane signal detect function

Change 124.5.5 as follows:

Various implementations of the Signal Detect function are permitted by this standard. When the MDIO is implemented, each PMD_signal_detect_*i*, where *i* represents the lane number in the range 0:~~3~~*n*–1, shall be continuously set in response to the magnitude of the optical signal on its associated lane, according to the requirements of Table 124–4.

124.5.8 PMD lane-by-lane transmit disable function (optional)

Change item a) in the lettered list in 124.5.8 as follows:

- a) When a PMD_transmit_disable_*i* variable (where *i* represents the lane number in the range 0:~~3~~*n*–1) is set to one, this function shall turn off the optical transmitter associated with that variable so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 124–6.

124.6 Lane assignments

Change 124.6 as follows:

There are no lane assignments (within a group of transmit or receive lanes) for 400GBASE-DR4, ~~400GBASE-DR4-2~~, 800GBASE-DR8, or 800GBASE-DR8-2. While it is expected that a PMD will map electrical lane *i* to optical lane *i* and vice versa, there is no need to define the physical ordering of the lanes, as the PCS sublayer is capable of receiving the lanes in any arrangement. The positioning of transmit and receive lanes at the MDI is specified in 124.11.3.1.

Change the title and text of 124.7 as follows:

124.7 PMD to MDI optical specifications for 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2

The operating ranges for the 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2 PMDs ~~are~~ is defined in Table 124–5. A 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, or 800GBASE-DR8-2 compliant PMD operates on type B1.1, B1.3, or B6 ~~a~~ single-mode fibers according to the specifications defined in Table 124–11. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 400GBASE-DR4 PMD operating at 600 m meets the operating range requirement of 2 m to 500 m).

Change Table 124–5 as follows:

Table 124–5—400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2 operating ranges

PMD type	Required operating range
<u>400GBASE-DR4 and 800GBASE-DR8</u>	2 m to 500 m
<u>400GBASE-DR4-2 and 800GBASE-DR8-2</u>	<u>2 m to 2 km</u>

Change the title and text of 124.7.1 as follows:

124.7.1 ~~400GBASE-DR4 transmitter~~ Transmitter optical specifications

~~The 400GBASE-DR4~~ A transmitter shall meet the specifications defined in Table 124–6 per the definitions in 124.8.

Replace Table 124–6 with the following table:

Table 124–6—400GBASE-DR4, 800GBASE-DR8, 400GBASE-DR4-2, and 800GBASE-DR8-2 transmit characteristics

Description	400GBASE-DR4	800GBASE-DR8	400GBASE-DR4-2 800GBASE-DR8-2	Unit
Signaling rate, each lane (range) 400GBASE-DR4, 400GBASE-DR4-2 800GBASE-DR8, 800GBASE-DR8-2	53.125 ± 100 ppm 53.125 ± 50 ppm			GBd GBd
Modulation format	PAM4			—
Lane wavelength (range)	1304.5 to 1317.5			nm
Side-mode suppression ratio (SMSR), (min)	30			dB
Average launch power, each lane (max)	4			dBm
Average launch power, each lane ^a (min)	−2.9 ^b			dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (max)	4.2			dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (min) ^c for TDECQ ≤ 3.4 dB for TDECQ < 1.4 dB for 1.4 dB ≤ TDECQ ≤ 3.4 dB	−0.8 — —	— −0.8 −2.2 + TDECQ	— −0.1 −1.5 + TDECQ	dBm dBm dBm
Launch power in OMA _{outer} minus TDECQ, each lane (min)	−2.2	—		dBm
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	3.4			dB
TDECQ − 10log ₁₀ (C _{eq}) ^d (max)	3.4	—		dB
Transmitter eye closure for PAM4 (TECQ) (max)	—	3.4		dB
TDECQ − TECQ (max)	—	2.5		dB
Transmitter overshoot and undershoot (max)	—	22		%
Transmitter power excursion (max)	—	5		dBm
Extinction ratio, each lane (min)	3.5			dB
Transmitter transition time (max)	17			ps
Average launch power of OFF transmitter, each lane (max)	−15			dBm

Table 124–6—400GBASE-DR4, 800GBASE-DR8, 400GBASE-DR4-2, and 800GBASE-DR8-2 transmit characteristics (*continued*)

Description	400GBASE-DR4	800GBASE-DR8	400GBASE-DR4-2 800GBASE-DR8-2	Unit
$RIN_{21.4OMA}$ (max)	−136			dB/Hz
Optical return loss tolerance (max)	21.4			dB
Transmitter reflectance ^c (max)	−26			dB

^a Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

^b Average launch power of −2.9 dBm corresponds to an OMA of −0.8 dBm with an extinction ratio of approximately 10 dB or an OMA of −0.1 dBm with an extinction ratio of approximately 16 dB.

^c For 400GBASE-DR4, the requirement on the OMA_{outer} (min) applies even in the cases where TDECQ < 1.4 dB.

^d C_{eq} is a coefficient defined in 121.8.5.3, which accounts for the reference equalizer noise enhancement.

^e Transmitter reflectance is defined looking into the transmitter.

Insert a new paragraph and Figure 124–2a at the end of 124.7.1 as follows:

For 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2, the values for OMA_{outer} (min) in Table 124–6 vary with TDECQ, as illustrated in Figure 124–2a along with the values for OMA_{outer} (max).

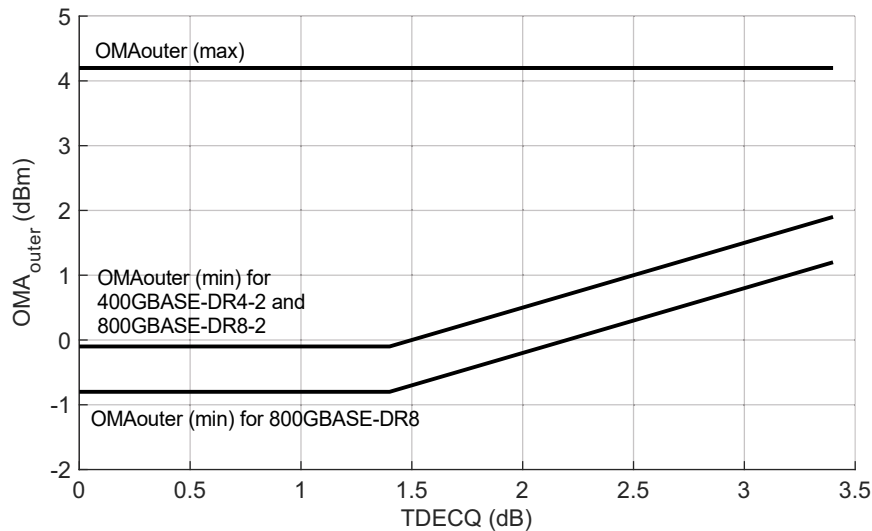


Figure 124–2a— OMA_{outer} (max) and OMA_{outer} (min) versus TDECQ for 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2

Change the title and text of 124.7.2 as follows:

124.7.2 400GBASE-DR4 ~~receive~~ Receive optical specifications

The ~~400GBASE-DR4~~ A receiver shall meet the specifications defined in Table 124–7 per the definitions in 124.8. ~~See~~ For 400GBASE-DR4 and 400GBASE-DR4-2, see NOTE at the end of 120.5.2 concerning the transition density of lanes operating at this nominal signaling rate.

Replace Table 124–7 with the following table:

Table 124–7—400GBASE-DR4, 800GBASE-DR8, 400GBASE-DR4-2, and 800GBASE-DR8-2 receive characteristics

Description	400GBASE-DR4	800GBASE-DR8	400GBASE-DR4-2 800GBASE-DR8-2	Unit
Signaling rate, each lane (range) 400GBASE-DR4, 400GBASE-DR4-2 800GBASE-DR8, 800GBASE-DR8-2	53.125 ± 100 ppm 53.125 ± 50 ppm			GBd GBd
Modulation format	PAM4			—
Lane wavelengths (range)	1304.5 to 1317.5			nm
Damage threshold ^a , each lane	5			dBm
Average receive power, each lane (max)	4			dBm
Average receive power, each lane ^b (min)	–5.9		–6.9	dBm
Receive power (OMA _{outer}), each lane (max)	4.2			dBm
Receiver reflectance (max)	–26			dB
Receiver sensitivity (OMA _{outer}), each lane (max) for SECQ ≤ 3.4 dB for TECQ < 1.4 dB for 1.4 dB ≤ TECQ ≤ 3.4 dB	Equation (124–1) ^c — —	— –3.9 –5.3 + TECQ	— –4.3 –5.7 + TECQ	dBm dBm dBm
Stressed receiver sensitivity (OMA _{outer}), each lane ^d (max)	–1.9			dBm

**Table 124–7—400GBASE-DR4, 800GBASE-DR8, 400GBASE-DR4-2, and 800GBASE-DR8-2
receive characteristics (*continued*)**

Description	400GBASE-DR4	800GBASE-DR8	400GBASE-DR4-2 800GBASE-DR8-2	Unit
Conditions of stressed receiver sensitivity test: ^e				
Stressed eye closure for PAM4 (SECQ), lane under test	3.4			dB
SECQ – 10log ₁₀ (C _{eq}) ^f (max), lane under test	3.4	—		dB
OMA _{outer} of each aggressor lane	4.2			dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c For 400GBASE-DR4, receiver sensitivity (OMA_{outer}), each lane (max) is optional and is defined for a transmitter with a value of SECQ up to 3.4 dB.

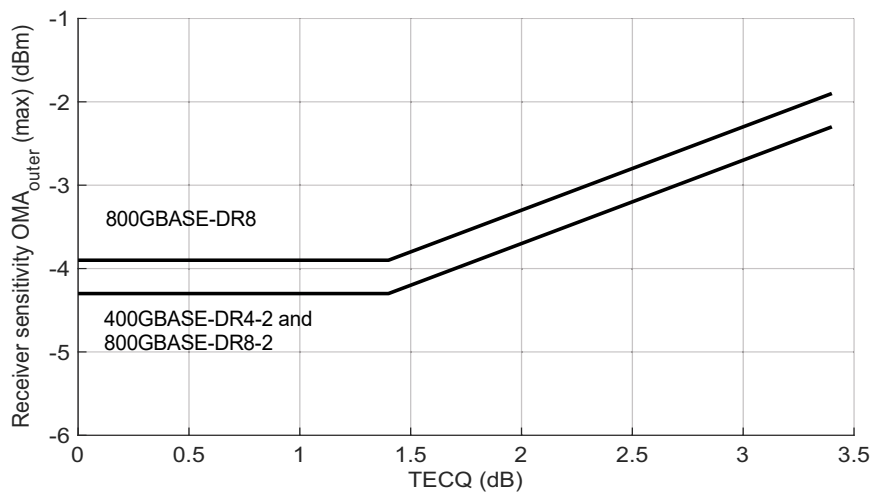
^d Measured with conformance test signal at TP3 (see 124.8.10) for the BER specified in 124.1.1.

^e These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

^f C_{eq} is a coefficient defined in 121.8.5.3, which accounts for the reference equalizer noise enhancement.

Insert a new paragraph and Figure 124–2b at the end of 124.7.2 as follows:

For 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2, the values for receiver sensitivity (OMA_{outer}) (max) in Table 124–7 vary with TECQ as illustrated in Figure 124–2b.



**Figure 124–2b—Receiver sensitivity (OMA_{outer}) (max) for 400GBASE-DR4-2,
800GBASE-DR8, and 800GBASE-DR8-2**

Change the title and text of 124.7.3 as follows:

124.7.3 ~~400GBASE-DR4 illustrative~~ Illustrative link power budget budgets

~~An illustrative~~ Illustrative power budgets and penalties for 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2 channels are shown in Table 124–8.

Change Table 124–8 as follows:

**Table 124–8—~~400GBASE-DR4, 800GBASE-DR8, 400GBASE-DR4-2, and 800GBASE-DR8-2~~
illustrative link power budgets**

Parameter	400GBASE-DR4 <u>800GBASE-DR8</u>	400GBASE-DR4-2 <u>800GBASE-DR8-2</u>	Unit
Power budget (for max TDECQ)	6.5	<u>7.6</u>	dB
Operating distance	500	<u>2000</u>	m
Channel insertion loss ^a	3	<u>4</u>	dB
Maximum discrete reflectance	See 124.11.2.2	<u>See 124.11.2.2</u>	dB
Allocation for penalties ^b (for max TDECQ)	3.5	<u>3.6</u>	dB
Additional insertion loss allowed	0	<u>0</u>	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 124–5 and cabled optical fiber attenuation of 0.5 dB/km at 1304.5 nm plus an allocation for connection and splice loss given in 124.11.2.1.

^b Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

Insert a new paragraph, Figure 124–2c, and Figure 124–2d at the end of 124.7.3 as follows:

The values of transmitter $\text{OMA}_{\text{outer}}$ (max), transmitter $\text{OMA}_{\text{outer}}$ (min) versus TDECQ, and receiver sensitivity ($\text{OMA}_{\text{outer}}$) (max) versus TECQ are illustrated in Figure 124–2c for 800GBASE-DR8, and in Figure 124–2d for 400GBASE-DR4-2 and 800GBASE-DR8-2.

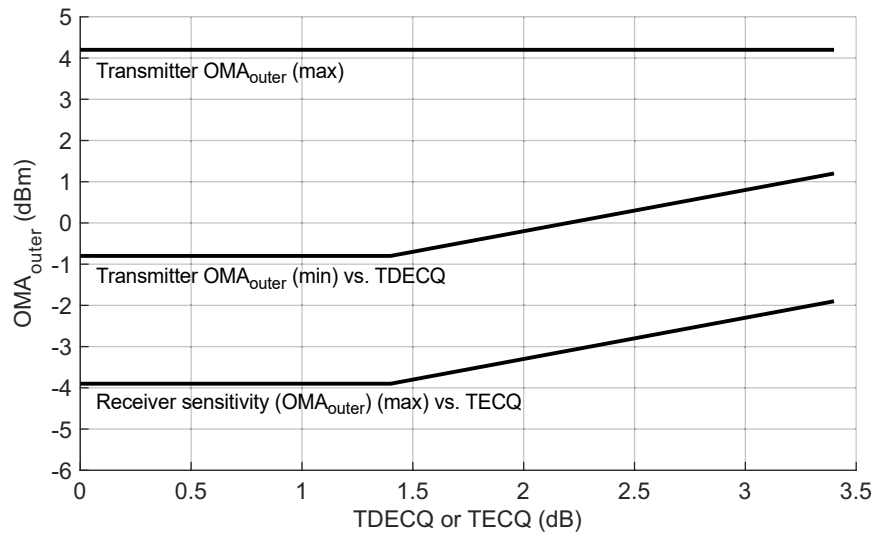


Figure 124-2c—Transmitter OMA_{outer} versus TDECQ and receiver sensitivity (OMA_{outer}) versus TECQ for 800GBASE-DR8

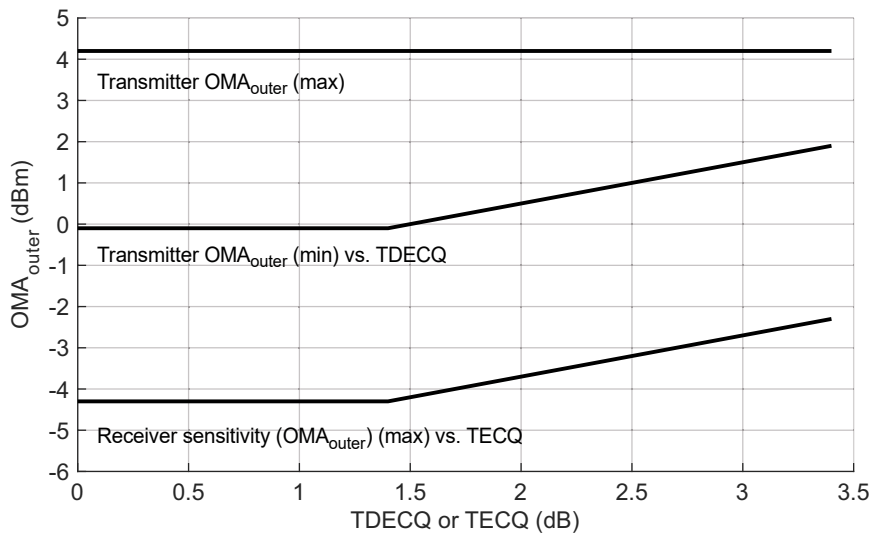


Figure 124-2d—Transmitter OMA_{outer} versus TDECQ and receiver sensitivity (OMA_{outer}) versus TECQ for 400GBASE-DR4-2 and 800GBASE-DR8-2

124.8 Definition of optical parameters and measurement methods

124.8.1 Test patterns for optical parameters

Change Table 124–9 as follows:

Table 124–9—Test patterns

Pattern	Pattern description	Defined in
Square wave	Square wave (8 threes, 8 zeros)	120.5.11.2.4
3	PRBS31Q	120.5.11.2.2
4	PRBS13Q	120.5.11.2.1
5	Scrambled idle	119.2.4.9 or 172.2.4.11
6	SSPRQ	120.5.11.2.3

Change Table 124–10 as follows:

Table 124–10—Test-pattern definitions and related subclauses

Parameter	Pattern	Related subclause
Wavelength	Square wave, 3, 4, 5, 6 _a or valid 400GBASE-R or 800GBASE-R signal	124.8.2
Side mode suppression ratio	3, 5, 6 _a or valid 400GBASE-R or 800GBASE-R signal	124.8.2
Average optical power	3, 5, 6 _a or valid 400GBASE-R or 800GBASE-R signal	124.8.3
Outer Optical Modulation Amplitude (OMA _{outer})	4 or 6	124.8.4
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	124.8.5
<u>Transmitter eye closure for PAM4 (TECQ)</u>	<u>6</u>	<u>124.8.5a</u>
<u>Transmitter overshoot and undershoot</u>	<u>6</u>	<u>124.8.5b</u>
<u>Transmitter power excursion</u>	<u>6</u>	<u>124.8.5c</u>
Extinction ratio	4 or 6	124.8.6
Transmitter transition time	Square wave or 6	124.8.7
RIN _{21.4} OMA	Square wave	124.8.8
<u>Receiver sensitivity</u>	<u>3 or 5</u>	<u>124.8.9</u>
Stressed receiver conformance test signal calibration	6	124.8.10
Stressed receiver sensitivity	3 or 5	124.8.10

124.8.5 Transmitter and dispersion eye closure for PAM4 (TDECQ)

Change 124.8.5 as follows:

The TDECQ and, for 400GBASE-DR4 only, $TDECQ - 10\log_{10}(C_{eq})$ of each lane shall be within the limits given in Table 124–6 if measured using the methods specified in 121.8.5.1, 121.8.5.2, and 121.8.5.3 using a reference equalizer as described in 121.8.5.4 where T is the symbol period for 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2, with the following exceptions:

- The signaling rate of the test pattern generator is as given in Table 124–6 and uses the test pattern specified for TDECQ in Table 124–10.
- The combination of the O/E converter and the oscilloscope has a 3 dB bandwidth of approximately 26.5625 GHz with a fourth-order Bessel-Thomson response to at least 1.3×53.125 GHz, and at frequencies above 1.3×53.125 GHz, the response should not exceed –20 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.
- The normalized noise power density spectrum $N(f)$ is equivalent to white noise filtered by a fourth-order Bessel-Thomson response filter with a 3 dB bandwidth of 26.5625 GHz.
- The 400GBASE-DR4-2 or 800GBASE-DR8-2 transmitter is tested using an optical channel as specified in 124.8.5.1.

Insert new subclause 124.8.5.1 at the end of 124.8.5 as follows:

124.8.5.1 Channel requirements

The 400GBASE-DR4-2 or 800GBASE-DR8-2 transmitter is tested using an optical channel that meets the requirements listed in Table 124–10a.

Table 124–10a—Transmitter compliance channel specifications

PMD type	Dispersion ^a (ps/nm)		Insertion loss ^b	Optical return loss ^c	Max mean DGD
	Minimum	Maximum			
400GBASE-DR4-2 or 800GBASE-DR4-2	$0.046 \times \lambda \times [1 - (1324 / \lambda)^4]$	$0.046 \times \lambda \times [1 - (1300 / \lambda)^4]$	Minimum	21.4 dB	0.8 ps

^a The dispersion is measured for the wavelength of the transmitter lane under test (λ in nm). The coefficient assumes 2 km for 400GBASE-DR4-2 or 800GBASE-DR8-2.

^b There is no intent to stress the sensitivity of the O/E converter associated with the oscilloscope.

^c The optical return loss is applied at TP2.

A 400GBASE-DR4-2 or 800GBASE-DR8-2 transmitter is to be compliant with a total dispersion at least as negative as the “minimum dispersion” and at least as positive as the “maximum dispersion” columns specified in Table 124–10a for the wavelength of the transmitter lane under test. This may be achieved with channels consisting of fibers with lengths chosen to meet the dispersion requirements.

To verify that the fiber has the correct amount of dispersion, the measurement method defined in IEC 60793-1-42 may be used. The measurement is made in the linear power regime of the fiber.

The channel provides an optical return loss specified in Table 124–10a. The state of polarization of the back reflection is adjusted to create the greatest RIN.

The mean DGD of the channel is to be less than the value specified in Table 124–10a.

Insert new subclauses 124.8.5a, 124.8.5b, and 124.8.5c after 124.8.5 as follows:

124.8.5a Transmitter eye closure for PAM4 (TECQ)

The transmitter eye closure for PAM4 (TECQ) is a measure of the optical transmitter's eye closure at TP2. For 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2, TECQ shall be within the limits given in Table 124–6 if measured using a test pattern specified for TECQ in Table 124–10. TECQ is measured using the methods specified for TDECQ in 124.8.5, except that the test fiber is not used.

124.8.5b Transmitter overshoot and undershoot

For 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2, the overshoot and undershoot shall be within the limits given in Table 124–6 if measured using a test pattern specified for overshoot and undershoot in Table 124–10.

Overshoot and undershoot are measured using the waveforms captured for the TDECQ test (see 124.8.5) and the waveform captured for the TECQ test (see 124.8.5a), but without the reference equalizer being applied in either case.

Overshoot and undershoot are defined in 140.7.7.

124.8.5c Transmitter power excursion

For 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2, the transmitter power excursion shall be within the limits given in Table 124–6 if measured using a test pattern specified for transmitter power excursion in Table 124–10.

Transmitter power excursion is measured using the waveforms captured for the TECQ test (see 124.8.5a), but without the reference equalizer being applied.

Transmitter power excursion is defined in 140.7.8.

124.8.9 Receiver sensitivity

Insert a new subclause heading 124.8.9.1 after the heading for 124.8.9 as follows:

124.8.9.1 Receiver sensitivity for 400GBASE-DR4

Change the text in 124.8.9.1 (formerly in 124.8.9) as follows (Figure 124–4 remains unchanged):

~~Receiver~~—The receiver sensitivity for 400GBASE-DR4 is optional and is defined for a transmitter with a value of SECQ up to 3.4 dB. Receiver sensitivity should meet Equation (124–1), which is illustrated in Figure 124–4.

$$RS = \max(-3.9, SECQ - 5.3) \quad (\text{dBm}) \quad (124-1)$$

where

RS is the receiver sensitivity
 $SECQ$ is the SECQ of the transmitter used to measure the receiver sensitivity

The normative requirement for 400GBASE-DR4 receivers is stressed receiver sensitivity.

Insert new subclause 124.8.9.2 after 124.8.9.1 as follows:

124.8.9.2 Receiver sensitivity for 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2

The receiver sensitivity ($\text{OMA}_{\text{outer}}$) for 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2 shall be within the limits given in Table 124–7 if measured using a test pattern for receiver sensitivity in Table 124–10. The conformance test signal at TP3 meets the requirements for a 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2 transmitter followed by an attenuator.

The TECQ of the conformance test signal is measured according to 124.8.5, except that the test fiber is not used. The measured value of TECQ is then used to calculate the limit for receiver sensitivity ($\text{OMA}_{\text{outer}}$) as specified in Table 124–7 for 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2.

124.9 Safety, installation, environment, and labeling

124.9.2 Laser safety

Change the first paragraph in 124.9.2 as follows:

400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2 optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

124.9.4 Environment

Change the first paragraph in 124.9.4 as follows:

Normative specifications in this clause shall be met by a system integrating a 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, or 800GBASE-DR8-2 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

124.9.5 Electromagnetic emission

Change 124.9.5 as follows:

A system integrating a 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, or 800GBASE-DR8-2 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

124.10 Fiber optic cabling model

Change Table 124–11 as follows:

Table 124–11—Fiber optic cabling (channel) characteristics

Description	400GBASE-DR4 800GBASE-DR8	<u>400GBASE-DR4-2</u> <u>800GBASE-DR8-2</u>	Unit
Operating distance (max)	500	<u>2000</u>	m
Channel insertion loss ^{a,b} (max)	3	<u>4</u>	dB
Channel insertion loss (min)	0	<u>0</u>	dB
Positive dispersion ^b (max)	0.8	<u>3.2</u>	ps/nm
Negative dispersion ^b (min)	−0.93	<u>−3.7</u>	ps/nm
DGD_max ^c	2.24	<u>2.3</u>	ps
Optical return loss (min)	37	<u>37</u>	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b Over the wavelength range 1304.5 nm to 1317.5 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

124.11 Characteristics of the fiber optic cabling (channel)

Change 124.11 as follows:

The ~~400GBASE-DR4~~ fiber optic cabling shall meet the specifications defined in [Table 124–12](#). The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together.

124.11.2 Optical fiber connection

124.11.2.1 Connection insertion loss

Change 124.11.2.1 as follows:

~~The~~ For 400GBASE-DR4 and 800GBASE-DR8 the maximum link distance is based on an allocation of 2.75 dB total connection and splice loss. For example, this allocation supports five connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 124–11 are met.

For 400GBASE-DR4-2 and 800GBASE-DR8-2 the maximum link distance is based on an allocation of 3 dB total connection and splice loss. For example, this allocation supports six connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 124–11 are met.

124.11.3 Medium Dependent Interface (MDI)

Change 124.11.3 as follows:

The ~~400GBASE-DR4~~ PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the “fiber optic cabling” (as shown in Figure 124-5). The ~~400GBASE-DR4~~ PMD is coupled to the fiber optic cabling through one connector plug into the MDI optical receptacle as shown in Figure 124-7. Example constructions of the MDI include the following:

- a) PMD with a connectorized fiber pigtail plugged into an adapter
- b) PMD receptacle

124.11.3.1 Optical lane assignments

Insert new subclause heading 124.11.3.1.1 after the heading 124.11.3.1 as follows:

124.11.3.1.1 Optical lane assignments for 400GBASE-DR4 and 400GBASE-DR4-2

Change 124.11.3.1.1 (formerly 124.11.3.1) and the title of Figure 124-6 as follows:

The four transmit and four receive optical lanes of 400GBASE-DR4 and 400GBASE-DR4-2 shall occupy the positions depicted in Figure 124-6 when looking into the MDI receptacle with the connector keyway feature on top. The interface contains eight active lanes within ~~twelve~~ 12 total positions. The transmit optical lanes occupy the left-most four positions. The receive optical lanes occupy the right-most four positions. The four center positions are unused.

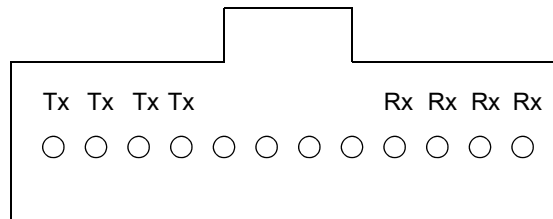


Figure 124-6—400GBASE-DR4 and 400GBASE-DR4-2 optical lane assignments

Insert new subclause 124.11.3.1.2, including Figure 124-6a, after Figure 124-6 as follows:

124.11.3.1.2 Optical lane assignments for 800GBASE-DR8 and 800GBASE-DR8-2

800GBASE-DR8 and 800GBASE-DR8-2 shall use a single-row 16-fiber interface. The eight transmit and eight receive optical lanes of 800GBASE-DR8 and 800GBASE-DR8-2 shall occupy the positions depicted in Figure 124-6a when looking into the MDI receptacle with the connector keyway feature on top. The interface contains 16 active lanes within 16 total positions. The transmit optical lanes occupy the leftmost eight positions. The receive optical lanes occupy the rightmost eight positions.

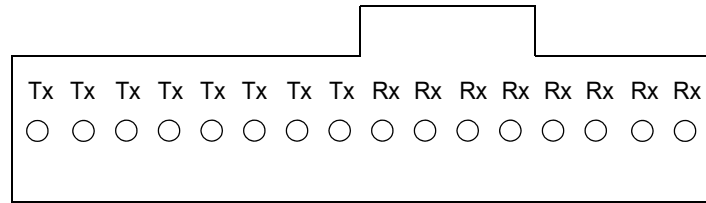


Figure 124–6a—Optical lane assignments for 800GBASE-DR8 and 800GBASE-DR8-2

Change the title of 124.11.3.2 as follows:

124.11.3.2 ~~Medium-Dependent Interface (MDI)~~ MDI requirements for 400GBASE-DR4 and 400GBASE-DR4-2

Insert new subclause 124.11.3.3 after 124.11.3.2 as follows:

124.11.3.3 MDI requirements for 800GBASE-DR8 and 800GBASE-DR8-2

The MDI shall optically mate with the compatible plug on the optical fiber cabling. The MDI shall meet the interface performance specifications of IEC 61753-021-2 for performance level D/2. 800GBASE-DR8 and 800GBASE-DR8-2 have a single-row 16-fiber interface optical lane assignment.

If the MDI is constructed with a connectorized fiber pigtail into an adapter, the connectorized pigtail shall meet the dimensional specifications of designation FOCIS 18 P-1x16-1-8-1-1-1 and the adapter shall meet the dimensional specifications of designation FOCIS 18 A-1-0 as defined in ANSI/TIA-604-18-A. If the MDI is constructed with a receptacle, it shall meet the dimensional specifications of designation FOCIS 18 R-1x16-1-8-1-1-2, as defined in ANSI/TIA-604-18-A. The plug terminating the optical fiber cabling that interconnects to either MDI configuration shall meet the dimensional specifications of designation FOCIS 18 P-1x16-1-8-2-1-1, as defined in ANSI/TIA-604-18-A. The MPO-16 female plug connector and MDI are structurally similar to those depicted in Figure 124–7, but with an angled end facet, 16 fibers, an offset keyway, and different pin diameters and locations.

Insert new subclause 124.11a after 124.11 as follows:

124.11a Interoperation between 400GBASE-DR4 and 400GBASE-DR4-2, and between 800GBASE-DR8 and 800GBASE-DR8-2

124.11a.1 Interoperation between 400GBASE-DR4 and 400GBASE-DR4-2

The 400GBASE-DR4 and 400GBASE-DR4-2 PMDs can interoperate with each other provided that the fiber optic cabling (channel) characteristics for 400GBASE-DR4 (see 124.10 and Table 124–11) are met.

124.11a.2 Interoperation between 800GBASE-DR8 and 800GBASE-DR8-2

The 800GBASE-DR8 and 800GBASE-DR8-2 PMDs can interoperate with each other provided that the fiber optic cabling (channel) characteristics for 800GBASE-DR8 (see 124.10 and Table 124–11) are met.

Change the title of 124.12 as follows:

**124.12 Protocol implementation conformance statement (PICS) proforma for
Clause 124, Physical Medium Dependent (PMD) sublayer and medium, type
400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2⁷**

124.12.1 Introduction

Change the first paragraph in 124.12.1 as follows:

The supplier of a protocol implementation that is claimed to conform to Clause 124, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2, shall complete the following protocol implementation conformance statement (PICS) proforma.

124.12.2 Identification

124.12.2.2 Protocol summary

Change the table in 124.12.2.2 as follows:

Identification of protocol standard	IEEE Std 802.3df-2024, Clause 124, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4, <u>400GBASE-DR4-2</u> , 800GBASE-DR8, and <u>800GBASE-DR8-2</u>
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-2024.)	

Date of Statement	
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⁷Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

124.12.3 Major capabilities/options

Insert four new rows at the start of the table in 124.12.3 as follows (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
*DR4	400GBASE-DR4 PMD	124.7	Device supports requirements for 400GBASE-DR4 PHY	O.1	Yes [] No []
*DR42	400GBASE-DR4-2 PMD	124.7	Device supports requirements for 400GBASE-DR4-2 PHY	O.1	Yes [] No []
*DR8	800GBASE-DR8 PMD	124.7	Device supports requirements for 800GBASE-DR8 PHY	O.1	Yes [] No []
*DR82	800GBASE-DR8-2 PMD	124.7	Device supports requirements for 800GBASE-DR8-2 PHY	O.1	Yes [] No []
...					

Change the title of 124.12.4 as follows:

124.12.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4, 400GBASE-DR4-2, 800GBASE-DR8, and 800GBASE-DR8-2

124.12.4.1 PMD functional specifications

Change the row for item “F1” in the table in 124.12.4.1 as follows (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 400GBASE-R _u or 800GBASE-R PCS and PMA	124.1		M	Yes []
...					

124.12.4.3 PMD to MDI optical specifications for 400GBASE-DR4

Change the table in 124.12.4.3 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
<u>DDR1</u>	Transmitter meets specifications in Table 124–6	124.7.1	Per definitions in 124.8	<u>DR4:M</u>	Yes [] <u>N/A</u> []
<u>DDR2</u>	Receiver meets specifications in Table 124–7	124.7.2	Per definitions in 124.8	<u>DR4:M</u>	Yes [] <u>N/A</u> []

Insert new subclauses 124.12.4.3a, 124.12.4.3b, and 124.12.4.3c after 124.12.4.3 as follows:

124.12.4.3a PMD to MDI optical specifications for 400GBASE-DR4-2

Item	Feature	Subclause	Value/Comment	Status	Support
EDR1	Transmitter meets specifications in Table 124–6	124.7.1	Per definitions in 124.8	DR42:M	Yes [] N/A []
EDR2	Receiver meets specifications in Table 124–7	124.7.2	Per definitions in 124.8	DR42:M	Yes [] N/A []

124.12.4.3b PMD to MDI optical specifications for 800GBASE-DR8

Item	Feature	Subclause	Value/Comment	Status	Support
FDR1	Transmitter meets specifications in Table 124–6	124.7.1	Per definitions in 124.8	DR8:M	Yes [] N/A []
FDR2	Receiver meets specifications in Table 124–7	124.7.2	Per definitions in 124.8	DR8:M	Yes [] N/A []

124.12.4.3c PMD to MDI optical specifications for 800GBASE-DR8-2

Item	Feature	Subclause	Value/Comment	Status	Support
GDR1	Transmitter meets specifications in Table 124–6	124.7.1	Per definitions in 124.8	DR82:M	Yes [] N/A []
GDR2	Receiver meets specifications in Table 124–7	124.7.2	Per definitions in 124.8	DR82:M	Yes [] N/A []

124.12.4.4 Optical measurement methods

Change the table in 124.12.4.4 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	124.8	2 m to 5 m in length	M	Yes []
OM2	Center wavelength and SMSR	124.8.2	Per IEC 61280-1-3 under modulated conditions	M	Yes []
OM3	Average optical power	124.8.3	Per IEC 61280-1-1	M	Yes []
OM4	OMA measurements	124.8.4	Each lane	M	Yes []
OM5	Transmitter and dispersion eye closure for PAM4 (TDECQ)	124.8.5	Each lane	M	Yes []
OM6	Extinction ratio	124.8.6	Each lane	M	Yes []
OM7	Transmitter transition time	124.8.7	Each lane	M	Yes []
OM8	Stressed receiver sensitivity	124.8.10	Each lane	M	Yes []
<u>OM9</u>	<u>Transmitter eye closure for PAM4 (TECQ)</u>	<u>124.8.5a</u>	<u>Each lane</u>	<u>DR42+</u> <u>DR8+</u> <u>DR82:M</u>	<u>Yes []</u> <u>N/A []</u>
<u>OM10</u>	<u>Transmitter overshoot and undershoot</u>	<u>124.8.5b</u>	<u>Each lane</u>	<u>DR42+</u> <u>DR8+</u> <u>DR82:M</u>	<u>Yes []</u> <u>N/A []</u>
<u>OM11</u>	<u>Transmitter power excursion</u>	<u>124.8.5c</u>	<u>Each lane</u>	<u>DR42+</u> <u>DR8+</u> <u>DR82:M</u>	<u>Yes []</u> <u>N/A []</u>
<u>OM12</u>	<u>Receiver sensitivity</u>	<u>124.8.9.2</u>	<u>Each lane</u>	<u>DR42+</u> <u>DR8+</u> <u>DR82:M</u>	<u>Yes []</u> <u>N/A []</u>

124.12.4.6 Characteristics of the fiber optic cabling and MDI

Change the table in 124.12.4.6 as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	124.11	Meets requirements specified in Table 124–11	INS:M	Yes [] N/A []
OC2	Maximum discrete reflectance	124.11.2.2	Meets requirements specified in Table 124–13	INS:M	Yes [] N/A []
OC3	MDI layout	124.11.3.1 124.11.3.1.1	Optical lane assignments per Figure 124–6	DR4+DR42:M	Yes [] N/A []
OC4	MDI dimensions	124.11.3.2	Per IEC 61754-7-1 interface 7-1-9	DR4+DR42:M	Yes [] N/A []
OC5	Cabling connector dimensions	124.11.3.2	Per IEC 61754-7-1 interface 7-1-1	(DR4+DR42)* INS:M	Yes [] N/A []
OC6	MDI mating	124.11.3.2	MDI optically mates with plug on the cabling	DR4+DR42:M	Yes [] N/A []
<u>OC7</u>	<u>MDI layout</u>	<u>124.11.3.1.2</u>	<u>Optical lane assignments per Figure 124–6a</u>	DR8+DR82:M	<u>Yes []</u> <u>N/A []</u>
<u>OC8</u>	<u>MDI mating</u>	<u>124.11.3.3</u>	<u>MDI optically mates with plug on the cabling, performance grade Bm/1m</u>	DR8+DR82:M	<u>Yes []</u> <u>N/A []</u>
<u>OC9</u>	<u>MDI dimensions</u>	<u>124.11.3.3</u>	<u>Per ANSI/TIA-604-18-A designation FOCIS 18 P-1x16-1-8-1-1-1 and designation FOCIS 18 A-1-0 or designation FOCIS 18 R-1x16-1-8-1-1-2</u>	DR8+DR82:M	<u>Yes []</u> <u>N/A []</u>
<u>OC10</u>	<u>Cabling connector dimensions</u>	<u>124.11.3.3</u>	<u>Per ANSI/TIA-604-18-A designation FOCIS 18 P-1x16-1-8-2-1-1</u>	(DR8+DR82)* INS:M	<u>Yes []</u> <u>N/A []</u>